

ORIGIN OF THE UNIVERSE

When and how did the universe begin? Was there a beginning? Perhaps it was always there. If there was a beginning, will there be an end? The study of the answers to these questions is called **cosmology**.

Following Edwin Hubble's discoveries about the expanding universe, two major theories about the beginning of the universe became popular — the **big bang theory** and the **steady state theory**.

1. The big bang ($t = 0$)

*It's hard to imagine, but at this moment there was no space and no time. All that existed was energy, which was concentrated into a single point called **singularity**.*

2. One ten million trillion trillion trillionths of a second later ($t = +\frac{1}{10^{43}}$ s)

Time and space had begun. Space was expanding quickly and the temperature was about 100 million trillion trillion degrees Celsius. (The current core temperature of the sun is 15 million degrees Celsius.)

3. One ten billion trillion trillionths of a second after the big bang ($t = +\frac{1}{10^{34}}$ s)

*The universe had expanded to about the size of a pea. Matter in the form of tiny particles such as electrons and **positrons** (positively charged electrons) had formed. Particles collided with each other, releasing huge amounts of energy in the form of light. Until this moment there was no light.*

4. One ten thousandth of a second after the big bang ($t = +\frac{1}{10^4}$ s)

Protons and neutrons had formed as a result of collisions between smaller particles. The universe was very bright because light was trapped as it was continually being reflected by particles.

5. One hundredth of a second after the big bang ($t = +\frac{1}{100}$ s)

The universe was still expanding and cooling rapidly. It had grown to the same size as our solar system, but there was still no such thing as an atom.

6. One second after the big bang ($t = +1$ s)

The universe was probably more than a trillion trillion kilometres across. It had cooled to about ten billion degrees Celsius.

7. Five minutes after the big bang ($t = +5$ min)

The nuclei of hydrogen, helium and lithium had formed among a sea of electrons.

8. Three hundred thousand years after the big bang ($t = +300\,000$ years)

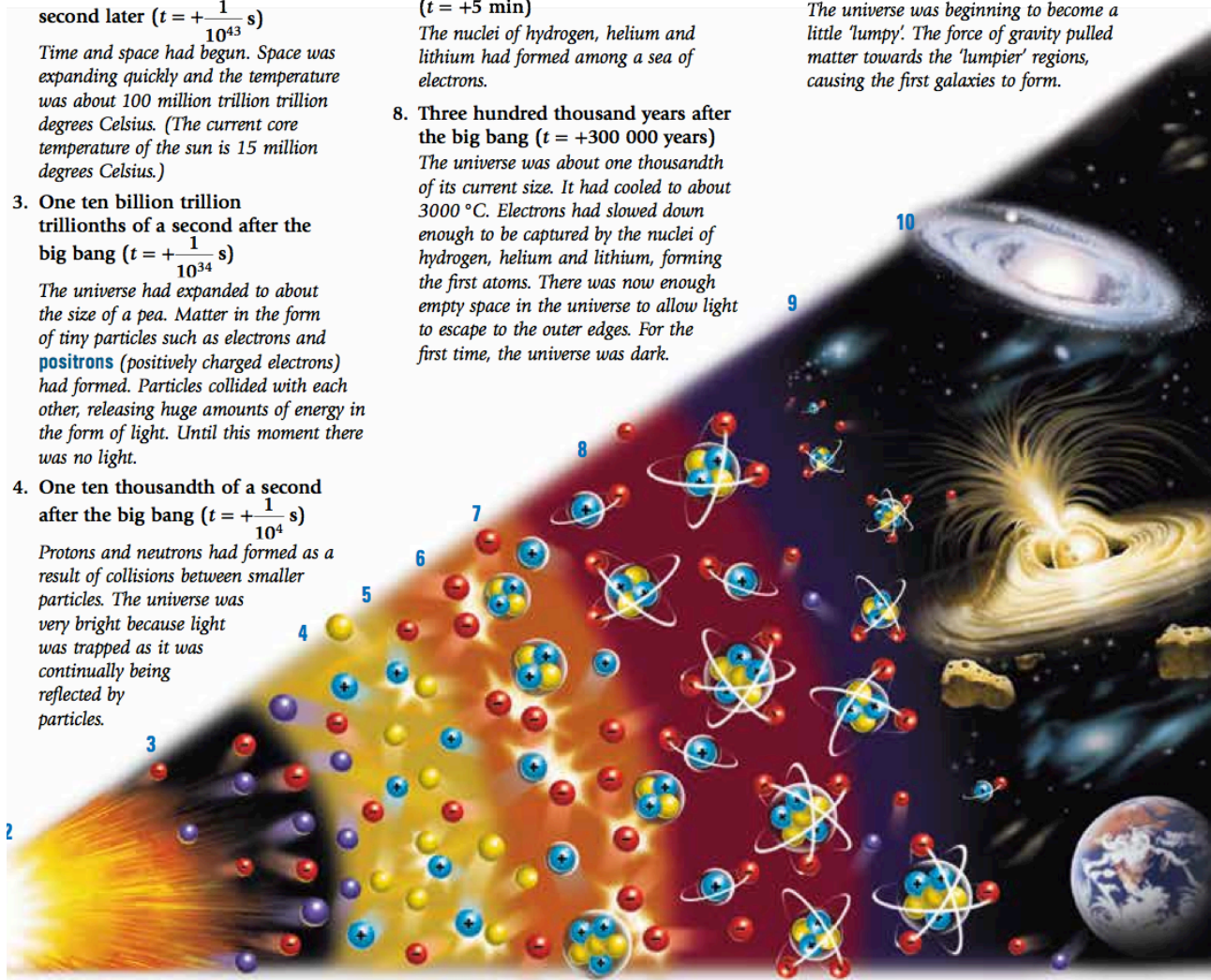
The universe was about one thousandth of its current size. It had cooled to about 3000°C . Electrons had slowed down enough to be captured by the nuclei of hydrogen, helium and lithium, forming the first atoms. There was now enough empty space in the universe to allow light to escape to the outer edges. For the first time, the universe was dark.

9. Two hundred million years after the big bang ($t = +200\,000\,000$ years)

*The first stars had appeared as gravity pulled atoms of hydrogen, helium and lithium together. **Nuclear reactions** took place inside the stars, causing the nuclei of the atoms to fuse together to form heavier nuclei. Around some of the newly forming stars, swirling clouds of matter cooled and formed clumps. This is how planets began to form.*

10. One billion years after the big bang ($t = +1\,000\,000\,000$ years)

The universe was beginning to become a little 'lumpy'. The force of gravity pulled matter towards the 'lumpier' regions, causing the first galaxies to form.



THE BIG BANG

According to the most commonly accepted theory among cosmologists, the universe began about 15 billion years ago with a 'big bang'. This is better described as "The Big Expansion".

At the beginning, all that existed was energy, which was concentrated into a single point called a **singularity**.

The Einstein connection

The singularity before the Big Bang was not 'nothing'. It was a huge amount of energy (with no mass) concentrated into a tiny, tiny point.

Einstein proposed that energy could be changed into matter. His equation $E = mc^2$ describes the change.

E represents the amount of energy in joules.

m represents the mass in kilograms.

c is the speed of light in metres per second (300 000 000 m/s).

Einstein's equation also describes how matter can be changed into energy. That is what happens in nuclear power stations, nuclear weapons and during the fusion process in stars.

HOW ABOUT THAT!

The big bang theory was first proposed in 1927 by Georges Lemaitre, a Catholic priest from Belgium. But it wasn't called the 'big bang theory' then. Ironically, the name 'big bang' was invented by Fred Hoyle, one of the developers of the steady state theory. He used the name to try to ridicule the cosmologists who proposed the big bang theory.

In 1929, Edwin Hubble's observations and law provided convincing evidence to support Lemaitre's proposal. In 1933, Lemaitre presented the details of his theory to an audience of scientists in California.

Albert Einstein, by then recognised as one of the greatest scientists of all time, was in the audience. At the end of Lemaitre's presentation Einstein stood, applauded and announced, 'That was the most beautiful and satisfactory explanation of creation that I have ever heard'.



The Steady State Theory

According to the Steady State Theory, which was proposed in 1948 by Fred Hoyle, there was no beginning of the universe. It was always there.

The galaxies are continually moving away from each other. In the extra space left between the galaxies, new stars and galaxies are created.

These new stars and galaxies replace those that move away, so that the universe always looks the same.

A huge debate between those who supported the Steady State Theory and those who supported the Big Bang Theory raged from 1948 until 1965. During that period, the evidence supporting the Big Bang Theory grew.

Supporting Evidence

- **The red shift**

The red shift provides evidence for an expanding universe. This evidence supports the Big Bang Theory and causes problems for those supporting the Steady State Theory.

A Steady State universe could expand only if new stars and galaxies replaced those that moved away. These young stars and galaxies have not been found by astronomers.

- **The elements**

The amounts of hydrogen and helium in the universe support the Big Bang Theory. The amount of helium in the universe is greater than that proposed by the Steady State theory, but it can be explained by its creation as a result of the Big Bang.

- **The afterglow**

When George Gamow and Ralph Alpher proposed their version of the Big Bang Theory in 1948, they calculated that the universe would now, about 15 billion years after creation, have a temperature of 2.7°C above **absolute zero** (about -270°C).

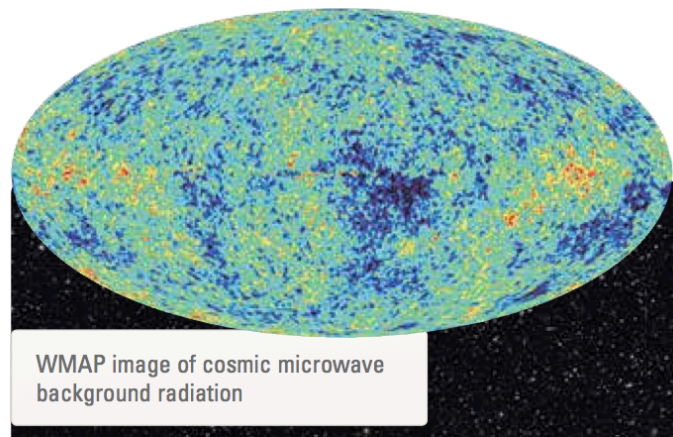
Gamow predicted that, because of its temperature, the universe would be emitting an 'afterglow' of radiation. This afterglow became known as cosmic microwave background radiation, which was discovered in 1965.

Its discovery put an end to the Steady State Theory, leaving the Big Bang Theory as the only theory supported by evidence currently available.

- **Cosmic Microwave Background Radiation**

The background radiation detected was released only 380 000 years after the Big Bang — the first radiation to escape.

The image shows how the temperature varied across the universe as it was 380 000 years after the Big Bang.



The blue parts of the map are the cooler regions. These regions were cool enough for atoms, and eventually galaxies, to form.

The red parts are warmer regions.

The map shows that galaxies are not evenly spread throughout the universe. They support the theory of an expanding universe that began with a Big Bang.

What is the future?

Will the expansion of the universe continue forever? If the universe does stop expanding, what will happen to it?

There are several competing theories about the answers to these questions.

One theory suggests that there is not enough mass in the universe for gravity to be able to pull it all back, so it will continue to expand forever.

Other theories suggest that the universe will eventually end. According to these theories, the end will come when:

- The universe snaps back onto itself in a 'big crunch' (the **Big Crunch Theory**). The end result will be a single point — singularity. Some cosmologists believe that the Big Crunch will be followed by another Big Bang.
- The expansion of the universe continues and stars use up their fuel and burn out, causing planets to freeze (the **Big Chill Theory**). The universe would then consist of scattered particles that never meet again.

- The universe rips itself apart violently as a result of expanding at an increasing speed (the ***Big Rip Theory***). According to this theory, the end of the universe will also be the end of time itself.